

Prediction model (in)stability

Clinical prediction models aiming to inform clinical decisions for individuals must be reliable if they should be implemented in clinical practise. Riley and Collins* explore the internal instability of example models developed using the Lasso on simulated data. The following plot was made using the paper's deposited code (Supplementary data), and emphasizes the volatility in individual predictions, particularly when the development data is small (which arguably is the most common scenario). Briefly, the plot shows the estimated risks from 1000 example models for nine individuals with known, true risks. Take-home message: Machine learning methods is not a remedy for small sample sizes, and internal model stability should be assessed when developing clinical prediction models.

